Cloud Computing and AWS Infrastructure

Cloud computing is the delivery of computing services including servers, storage, databases, networking, software, analytics, and intelligence over the internet to offer faster innovation, flexible resources, and economies of scale.

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally.

Key AWS Services

Amazon S3 (Simple Storage Service) is an object storage service offering industry-leading scalability, data availability, security, and performance. Customers use Amazon S3 to store and protect any amount of data for data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, and big data analytics.

AWS Lambda is a serverless compute service that lets you run code without provisioning or managing servers. Lambda runs your code only when needed and scales automatically, from a few requests per day to thousands per second. You pay only for the compute time you consume.

Amazon OpenSearch Service makes it easy to perform interactive log analytics, real-time application monitoring, website search, and more. OpenSearch is an open source, distributed search and analytics suite.

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models from leading AI companies like Anthropic, Cohere, Meta, Mistral AI, and Amazon through a single API, along with capabilities to build generative AI applications.

Infrastructure as Code

Terraform is an infrastructure as code tool that lets you define both cloud and on-premises resources in human-readable configuration files that you can version, reuse, and share. Terraform creates and manages resources on cloud platforms through their APIs.

Account Factory for Terraform (AFT) is a Terraform module maintained by AWS that follows a GitOps model to automate the provisioning and customization of AWS accounts. AFT provides a pipeline triggered when a Terraform configuration file is committed to a repository.

Security and Compliance

AWS Identity and Access Management (IAM) enables you to manage access to AWS services and resources securely. Using IAM, you can create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources.

AWS WAF is a web application firewall that helps protect your web applications or APIs against common web exploits and bots that may affect availability, compromise security, or consume excessive resources.

AWS Firewall Manager is a security management service which allows you to centrally configure and manage firewall rules across your accounts and applications in AWS Organizations.

MLOps and AI Infrastructure

MLOps is a set of practices that aims to deploy and maintain machine learning models in production reliably and efficiently. The MLOps lifecycle includes data preparation, model training, model evaluation, model deployment, and model monitoring.

A RAG (Retrieval Augmented Generation) system combines the power of large language models with the ability to retrieve relevant information from a knowledge base. Instead of relying solely on the model training data, RAG systems retrieve relevant documents and use them as context when generating responses.

Vector databases are specialized database systems designed to store and query high-dimensional vectors efficiently. They are commonly used in AI applications for semantic search, recommendation systems, and RAG pipelines. Examples include OpenSearch, Pinecone, and Weaviate.

Amazon Titan Embeddings is a text embeddings model that converts text into numerical representations called vectors that capture the semantic meaning of the text. These vectors are used for semantic search, text similarity, and other NLP tasks.

VPC and Networking

AWS VPC (Virtual Private Cloud) lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment.

VPC Endpoints allow you to privately connect your VPC to supported AWS services without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. Traffic between your VPC and the AWS service does not leave the Amazon network.